

Lecture 8: Oligopoly and Monopolistic Competition

Introduction to Economics

CAUGHT IN THE ACT



The law catches up with a colluding oligopolist.

What you will learn in this chapter:

- The meaning of **oligopoly**, and why it occurs
- How our understanding of oligopoly can be enhanced by using **game theory**, especially the concept of the **prisoners' dilemma**
- How repeated interactions among oligopolists can help them achieve **tacit collusion**
- How oligopoly works in practice, under the legal constraints of **antitrust policy**
- How prices and profits are determined in **monopolistic competition** in the short run and the long run
- Why oligopolists and monopolistically competitive firms differentiate their products
- The economic significance of advertising and **brand names**

Oligopoly

An **oligopoly** is an industry with only a small number of producers. A producer in such an industry is known as an **oligopolist**.

When no one firm has a monopoly, but producers nonetheless realize that they can affect market prices, an industry is characterized by **imperfect competition**.

An oligopoly consisting of only two firms is a **duopoly**. Each firm is known as a **duopolist**.

Sellers engage in **collusion** when they cooperate to raise each others' profits. A **cartel**, for example, OPEC (Organization of Petroleum Exporting Countries), is an agreement by several producers that increases their combined profits by telling each one how much to produce.

Oligopoly

Understanding Oligopoly

Making Cartel is illegal.

When the decisions of two or more firms significantly affect each others' profits, they are in a situation of **interdependence**.

The study of behavior in situations of interdependence is known as **game theory**.

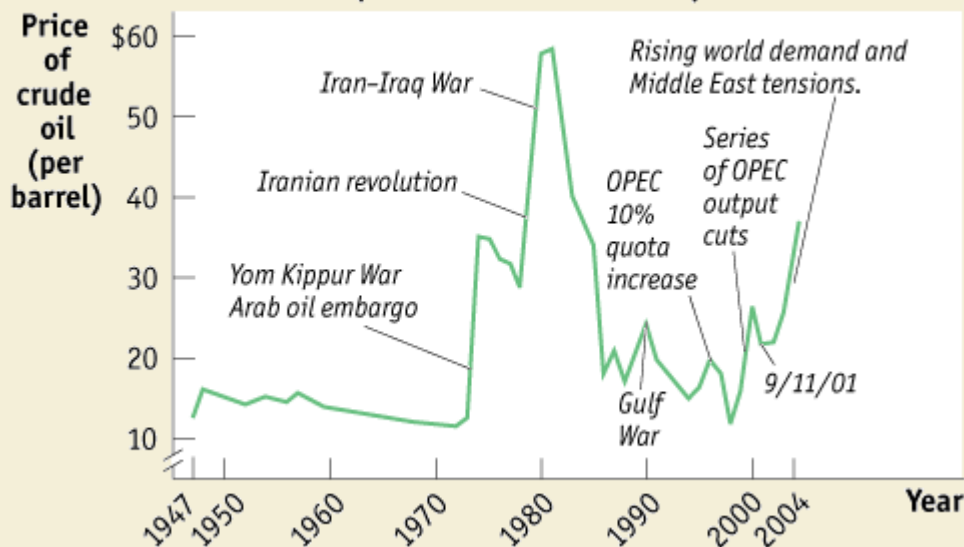
economics in action

The Rise and Fall and Rise of OPEC

Figure 12-4

The Ups and Downs of the Oil Cartel

Crude Oil Prices, 1947–2004
(in constant 2000 dollars)



The Organization of Petroleum Exporting Countries is a legal cartel that has had its ups and downs. From 1974 to 1985 it succeeded in driving world oil prices to unprecedented levels; then it collapsed. In 1998 the cartel once again became effective.

Source: WTRG Economics

WTI Spot price of Crude Oil



Oligopoly in Practice

Tacit Collusion and Price Wars

Large Numbers

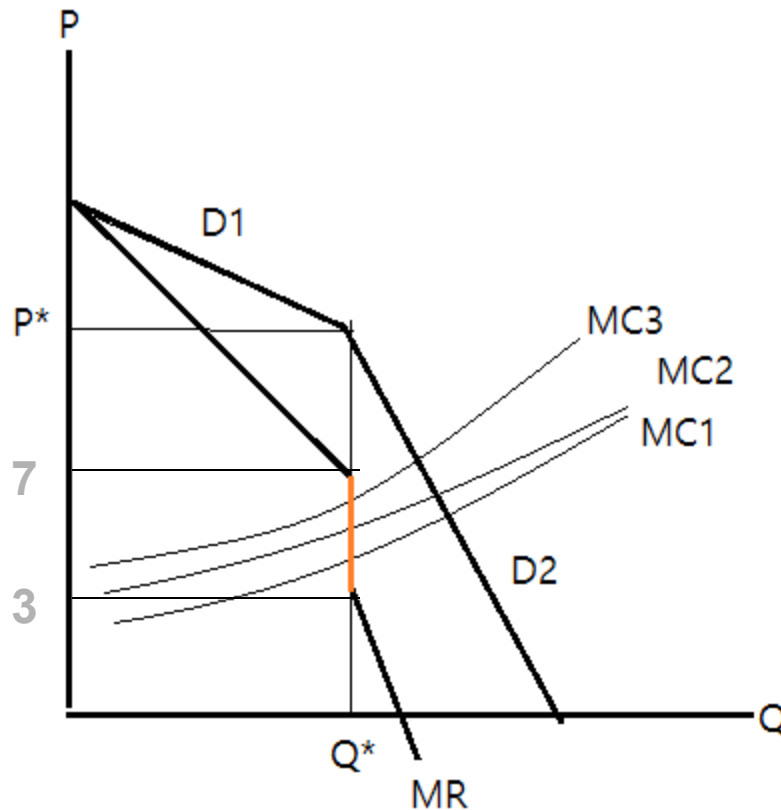
Complex Products and Pricing Schemes

Differences in Interests

Bargaining Power of Buyers

A **price war** occurs when tacit collusion breaks down and prices collapse.

Oligopoly and 'Kinked' Demand Curve



$$D1: p_1 = 11 - 0.5Q$$

$$D2: P_2 = 15 - 1.5Q$$

$$P^* = 9$$

$$Q^* = 4$$

Monopolistic Competition

Monopolistic competition is a market structure in which there are many competing producers in an industry, each producer sells a differentiated product, and there is free entry into and exit from the industry in the long run.

Large Numbers

Differentiated Products

Free Entry Into and Exit From the Industry in the Long Run

Monopolistic Products or Services

1. Large numbers
2. Free Entry and Exit in the Long Run
3. Differentiate Products by Style and Type, by Location, and by Quality

Examples :

Food Court, Gas Stations, Clothing Stores,
Books, Chocolate, Bicycles, Wines

Monopolistic Competition Market Differs from Other Types of Market

From Monopoly :

MC faces some competition

From Oligopoly:

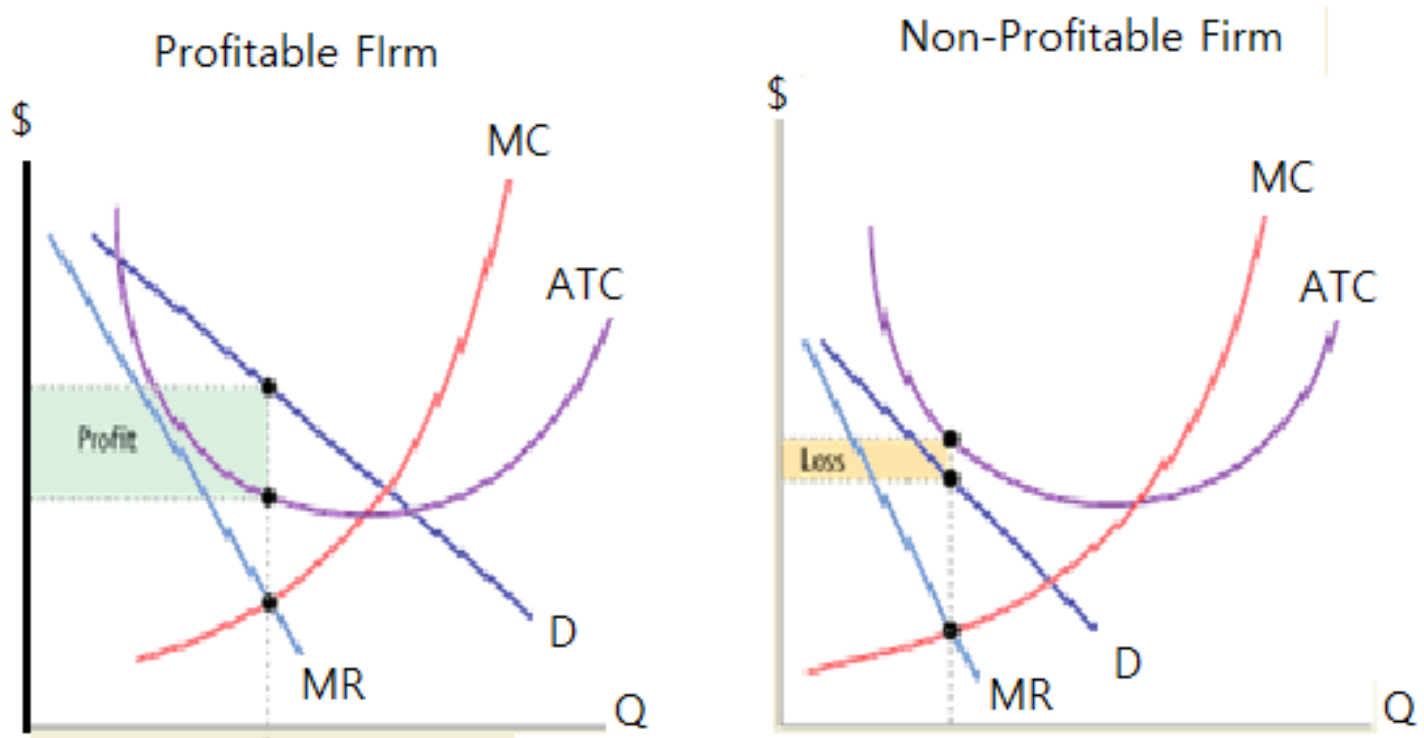
MC has many firms, free entry, and no incentives to collusion

From Perfect Competition :

MC has some market powers

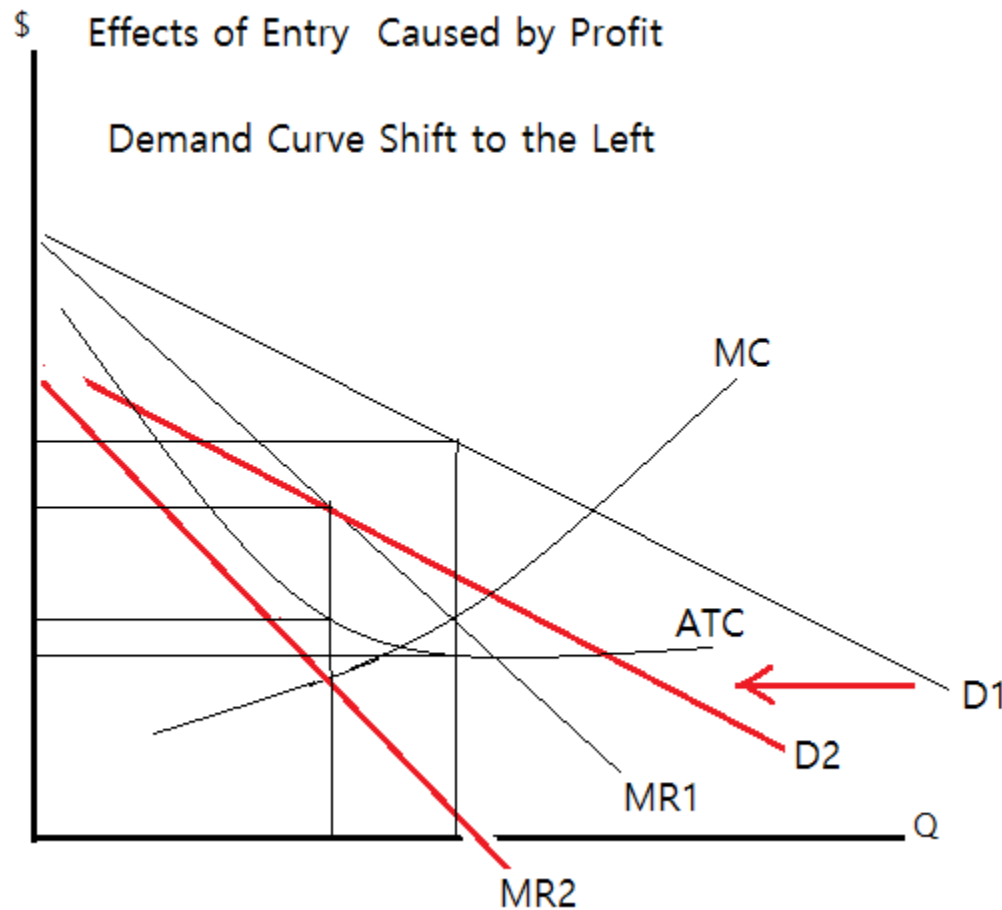
Monopolistic Competition

Monopolistic Competition in the Short Run



Monopolistic Competition

Monopolistic Competition in the Long Run



Monopolistic Competition

Monopolistic Competition in the Long Run

In the long run, a monopolistically competitive industry ends up in **zero-profit equilibrium**: each firm makes zero profit at its profit-maximizing quantity.

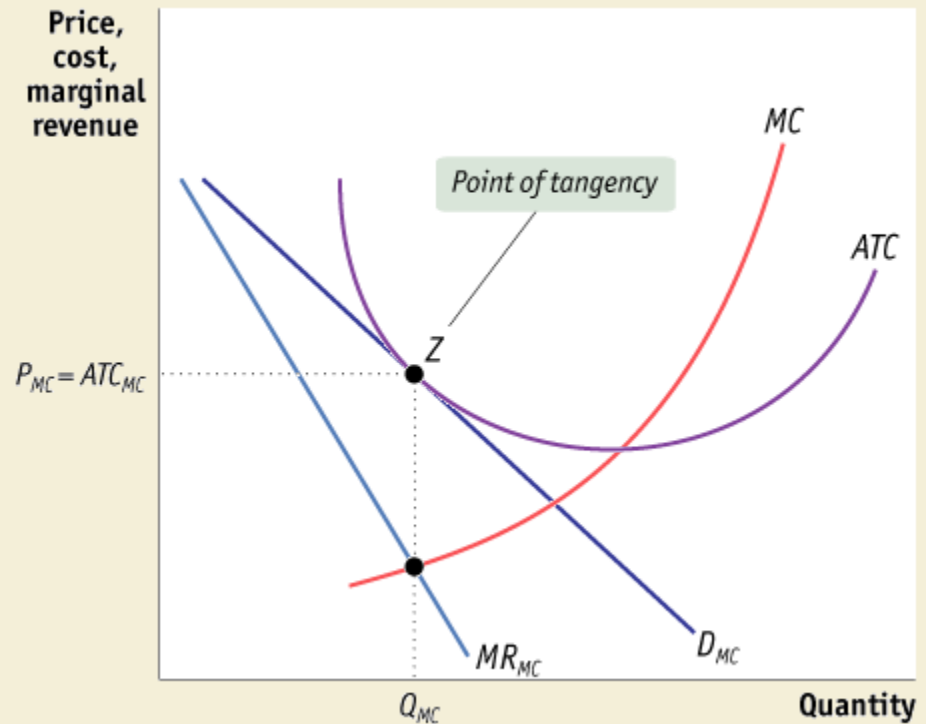
Monopolistic Competition

Monopolistic Competition in the Long Run

Figure 12-7

The Long-Run Zero-Profit Equilibrium

If existing firms are profitable, entry will occur and shift each existing firm's demand curve leftward. If existing firms are unprofitable, each remaining firm's demand curve shifts rightward as some firms exit the industry. Entry and exit will cease when every existing firm makes zero profit at its profit-maximizing quantity. So, in long-run zero-profit equilibrium, the demand curve of each firm is tangent to its average total cost curve at its profit-maximizing quantity: at the profit-maximizing quantity, Q_{MC} , price, P_{MC} , equals average total cost, ATC_{MC} . A monopolistically competitive firm is like a monopolist without monopoly profits.



Product Differentiation

Firms engage in **product differentiation** when they try to convince buyers that their product is different from the products of other firms in the industry.

There are three important forms of product differentiation:

- *differentiation by style or type*
- *differentiation by location*
- *differentiation by quality*

Product Differentiation

Controversies about Product Differentiation

The Role of Advertising

Advertising is worthwhile only in industries in which firms have at least some market power.

Given that advertising “works,” it’s not hard to see why firms with market power would spend money on it.

The big question about advertising is why it works. A related question is whether advertising is, from society’s point of view, a waste of resources.

Product Differentiation

Brand Names

A **brand name** is a name owned by a particular firm that distinguishes its products from those of other firms.

KEY TERMS

Oligopoly

Oligopolist

Imperfect competition

Duopoly

Duopolist

Collusion

Cartel

Interdependence

Game theory

Payoff

Payoff matrix

Prisoners' dilemma

Dominant strategy

Nash equilibrium

Noncooperative equilibrium

Strategic behavior

Tit for tat

Tacit collusion

Antitrust policy

Price war

Monopolistic competition

Zero-profit equilibrium

Product differentiation

Brand name